

King Faisal University
College of Computer Sciences and Information Technology
Computer Science Department

[CS-210]: [Programming Methodologies]
Section: []

Lab [06]: [While and Do-While Loops]

Date: []

Answer Sheet

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Computer Number:

Part 1 — Guided Debugging & Code Analysis

Error Identification & Correction Table

Fill in the table below by identifying the errors present in the code. One row is dedicated to each error.

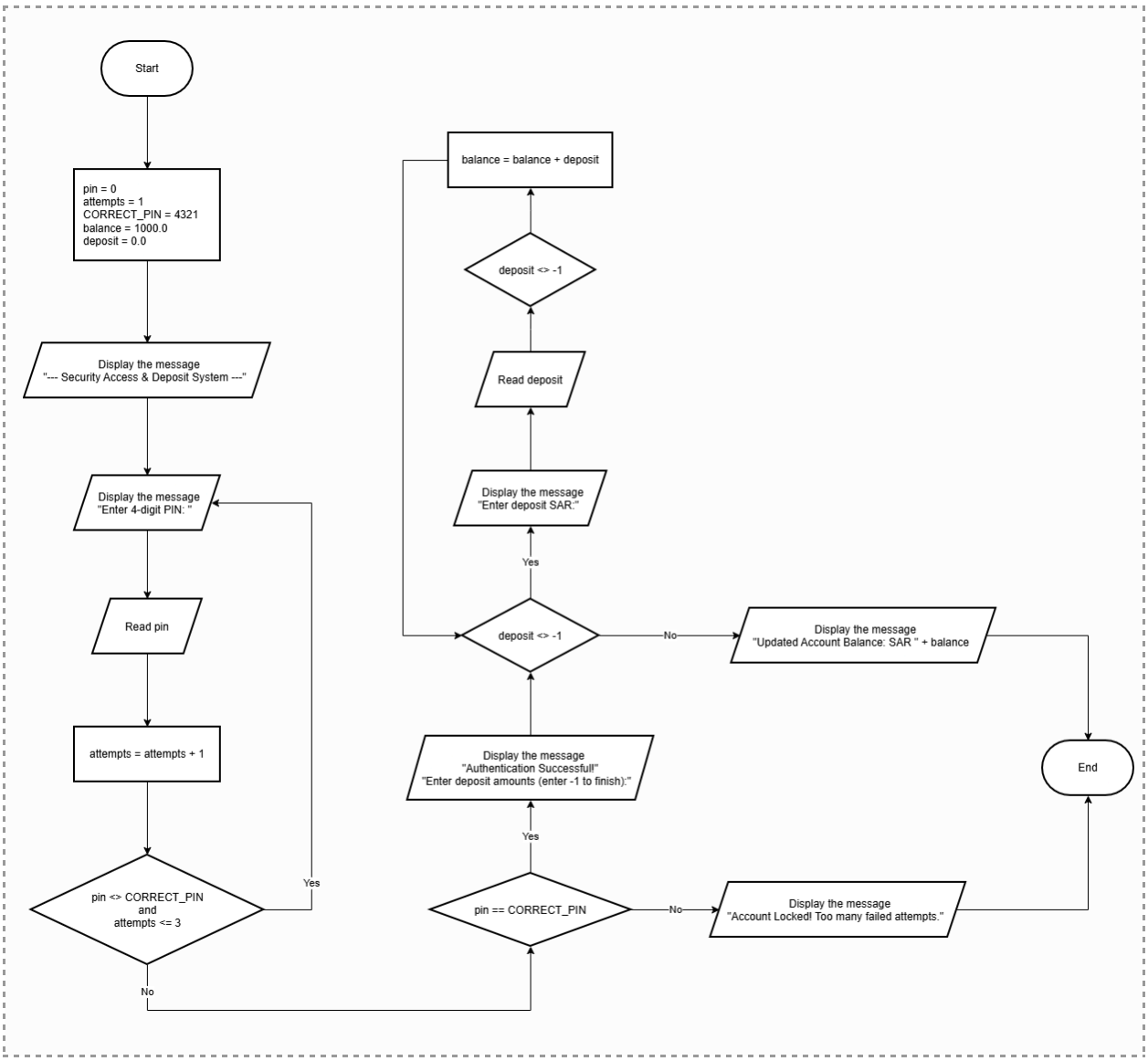
Line #	Code Snippet with Error	Error Description (Student Fill)	Error Type (Syntax/Logical)	Corrected Code (Student Fill)
19	} while (pin != CORRECT_PIN || attempts < 3)	Semicolon is forgotten	SyntaxError	} while (pin != CORRECT_PIN || attempts < 3);
19	} while (pin != CORRECT_PIN || attempts < 3);	Wrong condition	LogicalError	} while (pin != CORRECT_PIN && attempts < 3);
26	while (deposit != -1);	An additional semicolon closes the loop.	LogicalError	while (deposit != -1)

26	while (deposit != -1)	It includes only a single instruction. Instead, braces are needed; so all the belonging instructions will be executed within the loop.	LogicalError	while (deposit != -1){
				
				}

19	} while (pin != CORRECT_PIN && attempts < 3);	It only provides 2 attempts according to the condition attempts < 3. Thus, it should be attempts <= 3; To provide a third try.	LogicalError	} while (pin != CORRECT_PIN && attempts <= 3);

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



Corrected Source Code

Paste your complete, working, and corrected C++ source code in the box below.

```cpp
// Bank PIN Authentication & Balance Accumulator
#include <iostream>
using namespace std;

int main() {
    int pin = 0;
    int attempts = 1;
    const int CORRECT_PIN = 4321;
    double balance = 1000.0;
    double deposit = 0.0;

    cout << "--- Security Access & Deposit System ---" << endl;

    // PIN Verification using do-while loop
    do {
        cout << "Enter 4-digit PIN: ";
        cin >> pin;
        attempts++;
    } while (pin != CORRECT_PIN && attempts <= 3);

    if (pin == CORRECT_PIN) {
        cout << "Authentication Successful!" << endl;
        cout << "Enter deposit amounts (enter -1 to finish):" << endl;

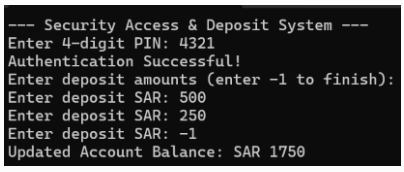
        // Sentinel-controlled deposit loop using while loop
        while (deposit != -1) {
            cout << "Enter deposit SAR: ";
            cin >> deposit;
            if (deposit != -1) {
                balance = balance + deposit;
            }
        }

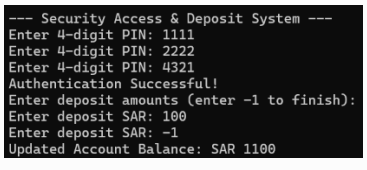
        cout << "Updated Account Balance: SAR " << balance << endl;
    }
    else {
        cout << "Account Locked! Too many failed attempts." << endl;
    }

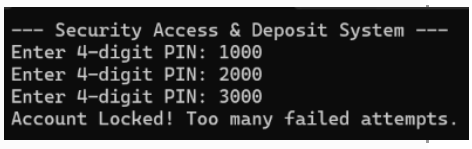
    return 0;
}
```

Execution Screenshot

Paste a screenshot of your working program output here showing a successful temperature conversion.



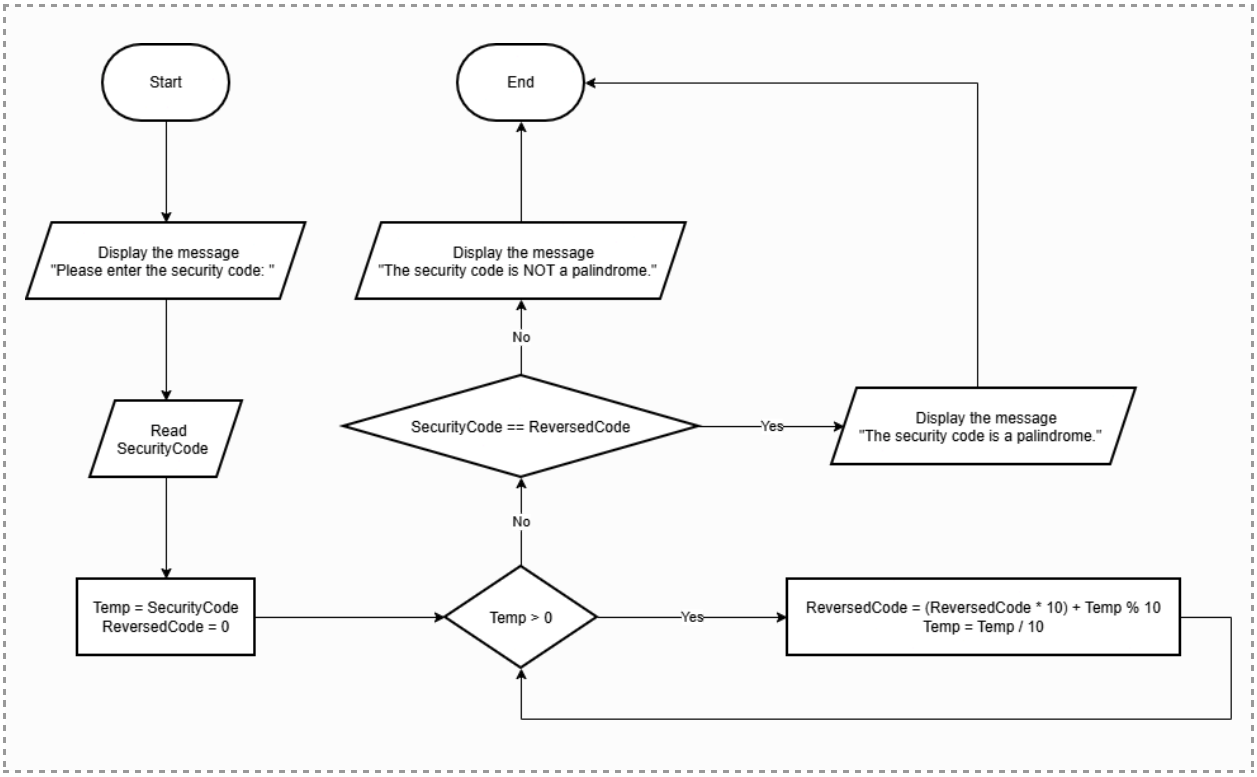




Part 2 — Independent Program Implementation

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



C++ Source Code:

Paste your complete C++ source code for the VAT calculator in the box below.

```cpp
#include <iostream>
using namespace std;

int main()
{
    // Unlike SecurityCode, ReversedCode's value won't be from the user.
    // That's why it needs to have an initializing value.
    int SecurityCode, ReversedCode = 0;

    cout << "Please enter the security code: ";
    cin >> SecurityCode;

    // that is a possible way.
    int Temp = SecurityCode;
    while (Temp > 0)
    {
        ReversedCode = (ReversedCode * 10) + Temp % 10;
        Temp /= 10;
    }

    if (SecurityCode == ReversedCode)
        cout << "The security code is a palindrome.\n";
    else
        cout << "The security code is NOT a palindrome.\n";

    return 0;
}
```

Execution Screenshot:

Paste a screenshot of your compiled VAT calculator running successfully with test data.

